

## cleaning

February 28, 2026

```
[7]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset
data = pd.read_csv("datasets/family_anxiety_14_dataset.csv") # training dataset
# data = pd.read_csv("datasets/enhanced_anxiety_dataset.csv") # testing dataset

# Display basic dataset information
print("DATASET DESCRIPTION")
print(f"Dataset shape: {data.shape[0]} rows, {data.shape[1]} columns")
print(f"Column names: {list(data.columns)}")
print(f>Data types:\n{data.dtypes}")
print(f"Missing values:\n{data.isnull().sum()}")

# Convert Yes/No columns to binary using map() instead of replace to avoid
↳warning
yes_no_cols = ['Smoking', 'Family History of Anxiety', 'Dizziness',
               'Medication', 'Recent Major Life Event']
for col in yes_no_cols:
    data[col] = data[col].map({'Yes': 1, 'No': 0})

# Creating target column: Diagnosed if Anxiety Level > 4
data['Diagnosed'] = (data['Anxiety Level (1-10)'] > 4).astype(int)

print("NEW TARGET COLUMN DIAGNOSED")
print(data[['Anxiety Level (1-10)', 'Diagnosed']].head())
print(f"Diagnosed distribution: 0 = {sum(data['Diagnosed']==0)}, 1 =
↳{sum(data['Diagnosed']==1)}")

# Features vs target
features = data.drop(columns=['Diagnosed'])
target = data['Diagnosed']

# Display min and max for numeric features after conversion
```

```

numeric_features = features.select_dtypes(include=[np.number])

print("NUMERIC FEATURES MIN AND MAX VALUES")
for col in numeric_features.columns:
    print(f"{col}: Min = {numeric_features[col].min()}, Max = {numeric_features[col].max()}")

# statistical summary
print("STATISTICAL SUMMARY OF NUMERIC FEATURES")
print(numeric_features.describe())

# OUTLIER DETECTION FOR CONTINUOUS COLUMNS
print("OUTLIER DETECTION FOR CONTINUOUS COLUMNS")

# Identify continuous columns
continuous_cols = ['Age', 'Anxiety Level (1-10)', 'Sleep Hours', 'Physical Activity (hrs/week)']

print(f"Continuous columns: {continuous_cols}")

# IQR Method for outlier detection
def detect_outliers_iqr(data, columns):
    outlier_summary = {}

    for col in columns:
        if col in data.columns:
            Q1 = data[col].quantile(0.25)
            Q3 = data[col].quantile(0.75)
            IQR = Q3 - Q1
            lower_bound = Q1 - 1.5 * IQR
            upper_bound = Q3 + 1.5 * IQR

            outliers = data[(data[col] < lower_bound) | (data[col] > upper_bound)]
            outlier_indices = outliers.index.tolist()

            outlier_summary[col] = {
                'Q1': Q1,
                'Q3': Q3,
                'IQR': IQR,
                'lower_bound': lower_bound,
                'upper_bound': upper_bound,
                'outlier_count': len(outliers),
                'outlier_percentage': (len(outliers) / len(data)) * 100,
                'outlier_indices': outlier_indices,
                'outlier_values': outliers[col].tolist()
            }

```

```

    return outlier_summary

# Z-Score Method for outlier detection
def detect_outliers_zscore(data, columns, threshold=3):
    outlier_summary = {}

    for col in columns:
        if col in data.columns:
            mean = data[col].mean()
            std = data[col].std()
            z_scores = np.abs((data[col] - mean) / std)

            outliers = data[z_scores > threshold]
            outlier_indices = outliers.index.tolist()

            outlier_summary[col] = {
                'mean': mean,
                'std': std,
                'threshold': threshold,
                'outlier_count': len(outliers),
                'outlier_percentage': (len(outliers) / len(data)) * 100,
                'outlier_indices': outlier_indices,
                'outlier_values': outliers[col].tolist(),
                'max_zscore': z_scores.max()
            }

    return outlier_summary

# Detect outliers using both methods
iqr_outliers = detect_outliers_iqr(data, continuous_cols)
zscore_outliers = detect_outliers_zscore(data, continuous_cols)

# Display IQR method results
print("IQR METHOD OUTLIER RESULTS")
for col, stats in iqr_outliers.items():
    print(f"{col}: {stats['outlier_count']} outliers, {stats['outlier_percentage']:.2f}%")
    print(f"    Bounds: [{stats['lower_bound']:.2f}, {stats['upper_bound']:.2f}]")
    if stats['outlier_count'] > 0:
        print(f"    Outlier values: {stats['outlier_values']}")

# Display Z-Score method results
print("ZSCORE METHOD OUTLIER RESULTS")
for col, stats in zscore_outliers.items():
    print(f"{col}: {stats['outlier_count']} outliers, {stats['outlier_percentage']:.2f}%")

```

```

print(f" Max ZScore: {stats['max_zscore']:.2f}")
if stats['outlier_count'] > 0:
    print(f" Outlier values: {stats['outlier_values']}")

# Box plots for visualizing outliers
fig, axes = plt.subplots(1, len(continuous_cols), figsize=(15, 5))
if len(continuous_cols) == 1:
    axes = [axes]

for i, col in enumerate(continuous_cols):
    axes[i].boxplot(data[col].dropna())
    axes[i].set_title(f'{col} Boxplot')
    axes[i].set_ylabel('Values')
plt.tight_layout()
plt.show()

# Numeric features correlation heatmap showing only high correlations
print("HIGH CORRELATIONS ABS > 0.2")
corr_matrix = numeric_features.corr(method='spearman')

# Create a mask for correlations with absolute value > 0.2
mask = np.abs(corr_matrix) > 0.2

# Remove diagonal
np.fill_diagonal(mask.values, False)

# Get pairs with high correlation
high_corr_pairs = []
for i in range(len(corr_matrix.columns)):
    for j in range(i+1, len(corr_matrix.columns)):
        if mask.iloc[i, j]:
            high_corr_pairs.append({
                'Feature 1': corr_matrix.columns[i],
                'Feature 2': corr_matrix.columns[j],
                'Correlation': corr_matrix.iloc[i, j]
            })

# Display high correlation pairs
if high_corr_pairs:
    high_corr_df = pd.DataFrame(high_corr_pairs)
    high_corr_df = high_corr_df.sort_values('Correlation', ascending=False)
    print(high_corr_df.to_string(index=False))
else:
    print("No correlations with absolute value > 0.2 found")

# Create heatmap only for high correlations
if high_corr_pairs:

```

```

plt.figure(figsize=(10, 8))

# Get unique features from high correlation pairs
high_corr_features = list(set([item['Feature 1'] for item in
↪high_corr_pairs] +
                               [item['Feature 2'] for item in
↪high_corr_pairs]))

# Filter correlation matrix for these features
high_corr_matrix = corr_matrix.loc[high_corr_features, high_corr_features]

sns.heatmap(high_corr_matrix, annot=True, fmt=".2f",
            cmap="coolwarm", center=0, linewidths=0.2, square=True)
plt.title("High Correlations Only Absolute > 0.2", fontsize=16,
↪fontweight='bold')
plt.tight_layout()
plt.show()

# 1. ENHANCED TARGET DISTRIBUTION PLOT - with percentages and clear labels
plt.figure(figsize=(10, 5))
ax = sns.countplot(x=data['Diagnosed'], hue=data['Diagnosed'],
                  palette=['skyblue', 'salmon'], legend=False)
plt.title("Diagnosed Target Distribution", fontsize=14, fontweight='bold')
plt.xlabel("Diagnosis Status (0 = Non-Diagnosed, 1 = Diagnosed)")
plt.ylabel("Count")

# Add count and percentage labels on bars
total = len(data)
for p in ax.patches:
    count = int(p.get_height())
    percentage = count/total*100
    ax.annotate(f'{count}\n({percentage:.1f}%)',
                (p.get_x() + p.get_width()/2., p.get_height()),
                ha='center', va='bottom', fontsize=11)
plt.tight_layout()
plt.show()

# 2. OVERLAYED HISTOGRAMS - diagnosed vs non-diagnosed with mean lines
numeric_cols = numeric_features.columns.tolist()
n_cols = len(numeric_cols)
n_rows = (n_cols + 1) // 2

fig, axes = plt.subplots(n_rows, 2, figsize=(14, 5*n_rows))
axes = axes.flatten()

for i, col in enumerate(numeric_cols):

```

```

ax = axes[i]

# Getting data for diagnosed and non-diagnosed separately
diagnosed_data = data[data['Diagnosed'] == 1][col]
nondiagnosed_data = data[data['Diagnosed'] == 0][col]

# Plotting histograms with transparency
ax.hist(nondiagnosed_data, bins=15, alpha=0.6, color='skyblue',
        label=f'Non-Diagnosed (n={len(nondiagnosed_data)})',
        edgecolor='black', linewidth=0.5)
ax.hist(diagnosed_data, bins=15, alpha=0.7, color='salmon',
        label=f'Diagnosed (n={len(diagnosed_data)})', edgecolor='black',
        linewidth=0.5)

# Adding vertical lines for means
ax.axvline(nondiagnosed_data.mean(), color='blue', linestyle='--',
           linewidth=2, label=f'Mean Non-Diagnosed: {nondiagnosed_data.
mean():.2f}')
ax.axvline(diagnosed_data.mean(), color='red', linestyle='--',
           linewidth=2, label=f'Mean Diagnosed: {diagnosed_data.mean():.
2f}')

ax.set_title(f'{col} Distribution by Diagnosis', fontsize=12,
fontweight='bold')
ax.set_xlabel(col)
ax.set_ylabel('Frequency')
ax.legend(loc='upper right', fontsize=8)
ax.grid(True, alpha=0.3)

# Hide any unused subplots
for j in range(i+1, len(axes)):
    axes[j].set_visible(False)

plt.tight_layout()
plt.show()

# 3. DENSITY PLOTS
fig, axes = plt.subplots(n_rows, 2, figsize=(14, 5*n_rows))
axes = axes.flatten()

for i, col in enumerate(numeric_cols):
    ax = axes[i]

    # Create density plots for each group
    for diagnosis, color, label in [(1, 'salmon', 'Diagnosed'), (0, 'skyblue',
        'Non-Diagnosed')]:

```

```

subset = data[data['Diagnosed'] == diagnosis]
sns.kdeplot(data=subset, x=col, ax=ax, color=color,
            label=f'{label} (n={len(subset)})',
            fill=True, alpha=0.4, linewidth=2)

ax.set_title(f'{col} Density by Diagnosis', fontsize=12, fontweight='bold')
ax.set_xlabel(col)
ax.set_ylabel('Density')
ax.legend(loc='upper right')
ax.grid(True, alpha=0.3)

# Hide any unused subplots
for j in range(i+1, len(axes)):
    axes[j].set_visible(False)

plt.tight_layout()
plt.show()

# Skewness and kurtosis for numeric features
skew_kurt = pd.DataFrame({
    'Skewness': numeric_features.skew(),
    'Kurtosis': numeric_features.kurtosis()
}).sort_values(by='Skewness', key=abs, ascending=False)

print("SKEWNESS AND KURTOSIS OF FEATURES")
print(skew_kurt)

# Categorical features summary
categorical_features = features.select_dtypes(include=['object', 'category']).
    columns

print("CATEGORICAL FEATURES SUMMARY")
for col in categorical_features:
    print(f"Categorical Feature: {col}")
    print(f"Number of unique values: {data[col].nunique()}")
    print("Value counts top 10:")
    print(data[col].value_counts().head(10))

# 4. PAIRED BAR CHART FOR BINARY FEATURES
print("\n" + "="*60)
print("BINARY FEATURES PERCENTAGES BY DIAGNOSIS GROUP")
print("="*60)

# Calculate percentages from your actual data
binary_cols = ['Smoking', 'Family History of Anxiety', 'Dizziness',
               'Medication', 'Recent Major Life Event']

```

```

# Feature names for display (cleaner labels)
feature_display_names = ['Smoking', 'Family History', 'Dizziness',
                        'Medication', 'Recent Life Event']

# Calculate percentages from the actual dataset
non_diagnosed_pcts = []
diagnosed_pcts = []

for col in binary_cols:
    diag_1_pct = data[data['Diagnosed'] == 1][col].mean() * 100
    diag_0_pct = data[data['Diagnosed'] == 0][col].mean() * 100
    non_diagnosed_pcts.append(diag_0_pct)
    diagnosed_pcts.append(diag_1_pct)

# Create the Paired Bar Chart
fig, ax = plt.subplots(figsize=(14, 8))

x = np.arange(len(feature_display_names))
width = 0.35

bars1 = ax.bar(x - width/2, non_diagnosed_pcts, width,
               label='Non-Diagnosed', color='skyblue', edgecolor='black',
               linewidth=0.5)
bars2 = ax.bar(x + width/2, diagnosed_pcts, width,
               label='Diagnosed', color='salmon', edgecolor='black',
               linewidth=0.5)

# Customize chart
ax.set_xlabel('Clinical Indicators', fontsize=14, fontweight='bold')
ax.set_ylabel('Percentage (%)', fontsize=14, fontweight='bold')
ax.set_title('Binary Clinical Indicators: Non-Diagnosed vs Diagnosed',
            fontsize=16, fontweight='bold', pad=20)
ax.set_xticks(x)
ax.set_xticklabels(feature_display_names, fontsize=12)
ax.legend(loc='upper right', fontsize=12, framealpha=0.9)
ax.set_ylim(0, 100)
ax.grid(True, alpha=0.3, axis='y', linestyle='--')

# Add value labels on bars
for bars in [bars1, bars2]:
    for bar in bars:
        height = bar.get_height()
        ax.text(bar.get_x() + bar.get_width()/2., height + 1,
                f'{height:.1f}%', ha='center', va='bottom',
                fontsize=11, fontweight='bold')

```

```

# Add a subtle background color for better readability
ax.set_facecolor('#f8f9fa')

plt.tight_layout()
plt.show()

# Print the percentages for reference
print("\nBinary Features - Percentageby Diagnosis Group:")
print("-" * 60)
for i, feature in enumerate(feature_display_names):
    print(f"{feature:20} Non-Diagnosed: {non_diagnosed_pcts[i]:.1f}% |  □
↳Diagnosed: {diagnosed_pcts[i]:.1f}% | Difference: {diagnosed_pcts[i] -
↳non_diagnosed_pcts[i]:.1f}%")

# Save cleaned dataset
data.to_csv("training_dataset.csv", index=False) # save the training dataset
# data.to_csv("testing_dataset.csv", index=False)
print("\n" + "="*60)
print("Dataset saved")
print(f"Final dataset shape: {data.shape[0]} rows, {data.shape[1]} columns")

```

#### DATASET DESCRIPTION

Dataset shape: 11000 rows, 19 columns

Column names: ['Age', 'Gender', 'Occupation', 'Sleep Hours', 'Physical Activity (hrs/week)', 'Caffeine Intake (mg/day)', 'Alcohol Consumption (drinks/week)', 'Smoking', 'Family History of Anxiety', 'Stress Level (1-10)', 'Heart Rate (bpm)', 'Breathing Rate (breaths/min)', 'Sweating Level (1-5)', 'Dizziness', 'Medication', 'Therapy Sessions (per month)', 'Recent Major Life Event', 'Diet Quality (1-10)', 'Anxiety Level (1-10)']

Data types:

|                                   |         |
|-----------------------------------|---------|
| Age                               | int64   |
| Gender                            | object  |
| Occupation                        | object  |
| Sleep Hours                       | float64 |
| Physical Activity (hrs/week)      | float64 |
| Caffeine Intake (mg/day)          | int64   |
| Alcohol Consumption (drinks/week) | int64   |
| Smoking                           | object  |
| Family History of Anxiety         | object  |
| Stress Level (1-10)               | int64   |
| Heart Rate (bpm)                  | int64   |
| Breathing Rate (breaths/min)      | int64   |
| Sweating Level (1-5)              | int64   |
| Dizziness                         | object  |
| Medication                        | object  |
| Therapy Sessions (per month)      | int64   |
| Recent Major Life Event           | object  |
| Diet Quality (1-10)               | int64   |

```

Anxiety Level (1-10)                float64
dtype: object
Missing values:
Age                                0
Gender                            0
Occupation                        0
Sleep Hours                       0
Physical Activity (hrs/week)       0
Caffeine Intake (mg/day)          0
Alcohol Consumption (drinks/week) 0
Smoking                           0
Family History of Anxiety          0
Stress Level (1-10)               0
Heart Rate (bpm)                  0
Breathing Rate (breaths/min)      0
Sweating Level (1-5)              0
Dizziness                         0
Medication                        0
Therapy Sessions (per month)      0
Recent Major Life Event            0
Diet Quality (1-10)               0
Anxiety Level (1-10)              0
dtype: int64
NEW TARGET COLUMN DIAGNOSED
  Anxiety Level (1-10)  Diagnosed
0                    3.0          0
1                    4.0          0
2                    6.0          1
3                    2.0          0
4                    4.0          0
Diagnosed distribution: 0 = 7075, 1 = 3925
NUMERIC FEATURES MIN AND MAX VALUES
Age: Min = 18, Max = 64
Sleep Hours: Min = 0.2, Max = 12.5
Physical Activity (hrs/week): Min = 0.0, Max = 11.2
Caffeine Intake (mg/day): Min = 0, Max = 599
Alcohol Consumption (drinks/week): Min = 0, Max = 19
Smoking: Min = 0, Max = 1
Family History of Anxiety: Min = 0, Max = 1
Stress Level (1-10): Min = 1, Max = 10
Heart Rate (bpm): Min = 60, Max = 119
Breathing Rate (breaths/min): Min = 12, Max = 29
Sweating Level (1-5): Min = 1, Max = 5
Dizziness: Min = 0, Max = 1
Medication: Min = 0, Max = 1
Therapy Sessions (per month): Min = 0, Max = 9
Recent Major Life Event: Min = 0, Max = 1
Diet Quality (1-10): Min = 1, Max = 10

```

Anxiety Level (1-10): Min = 1.0, Max = 10.0

# STATISTICAL SUMMARY OF NUMERIC FEATURES

|       | Age          | Sleep Hours  | Physical Activity (hrs/week) \ |
|-------|--------------|--------------|--------------------------------|
| count | 11000.000000 | 11000.000000 | 11000.000000                   |
| mean  | 40.414818    | 6.772427     | 2.916464                       |
| std   | 13.349472    | 1.615210     | 1.836844                       |
| min   | 18.000000    | 0.200000     | 0.000000                       |
| 25%   | 29.000000    | 5.600000     | 1.400000                       |
| 50%   | 40.000000    | 6.800000     | 2.800000                       |
| 75%   | 52.000000    | 7.900000     | 4.200000                       |
| max   | 64.000000    | 12.500000    | 11.200000                      |

|       | Caffeine Intake (mg/day) | Alcohol Consumption (drinks/week) \ |
|-------|--------------------------|-------------------------------------|
| count | 11000.000000             | 11000.000000                        |
| mean  | 313.291364               | 9.725545                            |
| std   | 172.920481               | 5.716079                            |
| min   | 0.000000                 | 0.000000                            |
| 25%   | 163.000000               | 5.000000                            |
| 50%   | 326.000000               | 10.000000                           |
| 75%   | 462.000000               | 15.000000                           |
| max   | 599.000000               | 19.000000                           |

|       | Smoking      | Family History of Anxiety | Stress Level (1-10) \ |
|-------|--------------|---------------------------|-----------------------|
| count | 11000.000000 | 11000.000000              | 11000.000000          |
| mean  | 0.226273     | 0.133545                  | 5.808545              |
| std   | 0.418437     | 0.340179                  | 2.941883              |
| min   | 0.000000     | 0.000000                  | 1.000000              |
| 25%   | 0.000000     | 0.000000                  | 3.000000              |
| 50%   | 0.000000     | 0.000000                  | 6.000000              |
| 75%   | 0.000000     | 0.000000                  | 8.000000              |
| max   | 1.000000     | 1.000000                  | 10.000000             |

|       | Heart Rate (bpm) | Breathing Rate (breaths/min) | Sweating Level (1-5) \ |
|-------|------------------|------------------------------|------------------------|
| count | 11000.000000     | 11000.000000                 | 11000.000000           |
| mean  | 90.994000        | 20.790909                    | 3.081273               |
| std   | 17.138795        | 5.181014                     | 1.401405               |
| min   | 60.000000        | 12.000000                    | 1.000000               |
| 25%   | 76.000000        | 16.000000                    | 2.000000               |
| 50%   | 92.000000        | 21.000000                    | 3.000000               |
| 75%   | 106.000000       | 25.000000                    | 4.000000               |
| max   | 119.000000       | 29.000000                    | 5.000000               |

|       | Dizziness    | Medication   | Therapy Sessions (per month) \ |
|-------|--------------|--------------|--------------------------------|
| count | 11000.000000 | 11000.000000 | 11000.000000                   |
| mean  | 0.517909     | 0.508818     | 4.684727                       |
| std   | 0.499702     | 0.499945     | 2.852922                       |
| min   | 0.000000     | 0.000000     | 0.000000                       |
| 25%   | 0.000000     | 0.000000     | 2.000000                       |

|     |          |          |          |
|-----|----------|----------|----------|
| 50% | 1.000000 | 1.000000 | 5.000000 |
| 75% | 1.000000 | 1.000000 | 7.000000 |
| max | 1.000000 | 1.000000 | 9.000000 |

|       | Recent Major Life Event | Diet Quality (1-10) | Anxiety Level (1-10) |
|-------|-------------------------|---------------------|----------------------|
| count | 11000.000000            | 11000.000000        | 11000.000000         |
| mean  | 0.522818                | 5.212727            | 4.149455             |
| std   | 0.499502                | 2.879370            | 2.115930             |
| min   | 0.000000                | 1.000000            | 1.000000             |
| 25%   | 0.000000                | 3.000000            | 3.000000             |
| 50%   | 1.000000                | 5.000000            | 4.000000             |
| 75%   | 1.000000                | 8.000000            | 5.000000             |
| max   | 1.000000                | 10.000000           | 10.000000            |

## OUTLIER DETECTION FOR CONTINUOUS COLUMNS

Continuous columns: ['Age', 'Anxiety Level (1-10)', 'Sleep Hours', 'Physical Activity (hrs/week)']

## IQR METHOD OUTLIER RESULTS

Age: 0 outliers (0.00%)

Bounds: [-5.50, 86.50]

Anxiety Level (1-10): 694 outliers (6.31%)

Bounds: [0.00, 8.00]

[illegible]

9.0, 9.0, 10.0, 10.0, 10.0, 9.0, 10.0, 10.0, 9.0, 10.0, 9.0, 10.0, 10.0, 10.0,  
10.0, 9.0, 9.0, 10.0, 10.0, 10.0, 9.0, 9.0, 9.0, 9.0, 10.0, 9.0, 9.0, 9.0, 9.0,  
9.0, 9.0, 10.0, 9.0, 9.0, 9.0, 9.0, 9.0, 9.0, 10.0, 9.0, 10.0, 9.0, 10.0, 10.0,  
9.0, 9.0, 9.0, 9.0, 10.0, 9.0, 9.0, 9.0, 10.0, 9.0, 9.0, 9.0, 10.0, 10.0, 9.0,  
10.0, 9.0, 10.0, 10.0, 9.0, 10.0, 9.0, 10.0, 9.0, 10.0, 9.0, 9.0, 9.0, 9.0,  
10.0, 10.0, 10.0, 9.0, 9.0, 10.0, 10.0, 10.0, 10.0, 9.0, 10.0, 9.0, 9.0, 10.0,  
10.0, 10.0, 9.0, 10.0, 9.0, 9.0, 10.0, 10.0, 9.0, 10.0, 9.0, 10.0, 9.0, 9.0,  
9.0, 9.0, 10.0, 10.0, 10.0, 9.0, 9.0, 9.0, 9.0, 10.0, 10.0, 9.0, 9.0, 10.0,  
10.0, 10.0, 10.0, 9.0, 10.0, 10.0, 10.0, 9.0, 9.0, 10.0, 9.0, 10.0, 9.0, 10.0,  
10.0, 9.0, 10.0, 9.0, 10.0, 9.0, 9.0, 9.0, 10.0, 9.0, 9.0, 9.0, 10.0, 10.0, 9.0,  
9.0, 9.0, 10.0, 9.0, 10.0, 9.0, 10.0, 10.0, 9.0, 9.0, 10.0, 9.0, 10.0, 10.0,  
10.0, 9.0, 9.0, 9.0, 9.0, 10.0, 10.0, 10.0, 10.0, 10.0, 10.0, 10.0, 10.0, 9.0,  
9.0, 10.0, 10.0, 10.0, 9.0, 10.0, 10.0, 9.0, 10.0, 10.0, 10.0, 9.0, 10.0, 9.0,  
9.0, 10.0, 9.0, 10.0, 9.0, 9.0, 9.0, 10.0, 9.0, 10.0, 9.0, 10.0, 9.0, 10.0, 9.0,  
10.0, 10.0, 9.0, 9.0, 9.0, 10.0, 9.0, 9.0, 9.0, 9.0, 10.0, 9.0, 9.0, 9.0, 10.0,  
10.0, 10.0, 9.0, 9.0, 10.0, 9.0, 9.0, 9.0, 9.0, 10.0, 9.0, 9.0, 9.0, 10.0, 9.0,  
10.0, 9.0, 9.0, 9.0, 9.0, 9.0, 9.0, 10.0, 9.0, 9.0, 9.0, 9.0, 9.0, 10.0, 10.0,  
10.0, 9.0, 10.0, 10.0, 9.0, 10.0, 9.0, 10.0, 10.0, 9.0, 10.0, 10.0, 9.0, 9.0,  
10.0, 9.0, 10.0, 9.0, 10.0, 9.0, 9.0, 10.0, 9.0, 9.0, 9.0, 9.0, 10.0, 10.0, 9.0,  
9.0, 10.0, 10.0, 9.0, 10.0, 9.0, 10.0, 10.0, 9.0, 10.0, 9.0, 10.0, 10.0, 10.0,  
9.0, 9.0, 10.0, 10.0, 10.0, 9.0, 9.0, 10.0, 9.0, 9.0, 9.0, 10.0, 9.0, 10.0, 9.0,  
9.0, 10.0, 10.0, 10.0, 9.0, 10.0, 9.0]

Sleep Hours: 28 outliers (0.25%)

Bounds: [2.15, 11.35]

Outlier values: [11.8, 11.4, 2.1, 11.6, 12.2, 11.7, 1.9, 11.4, 2.0, 12.5, 1.8,  
11.4, 11.4, 1.7, 11.5, 1.8, 1.6, 11.9, 11.9, 1.3, 12.1, 11.7, 0.2, 11.4, 11.4,  
12.5, 11.4, 11.7]

Physical Activity (hrs/week): 30 outliers (0.27%)

Bounds: [-2.80, 8.40]

Outlier values: [10.8, 9.2, 8.5, 9.8, 10.2, 8.6, 8.5, 8.5, 10.0, 9.3, 8.6,  
8.7, 9.6, 8.5, 9.5, 11.2, 9.8, 8.5, 9.0, 8.5, 9.4, 8.7, 9.9, 8.6, 9.1, 8.6, 9.0,  
8.9, 9.0, 8.9]

ZSCORE METHOD OUTLIER RESULTS

Age: 0 outliers (0.00%)

Max ZScore: 1.77

Anxiety Level (1-10): 0 outliers (0.00%)

Max ZScore: 2.76

Sleep Hours: 17 outliers (0.15%)

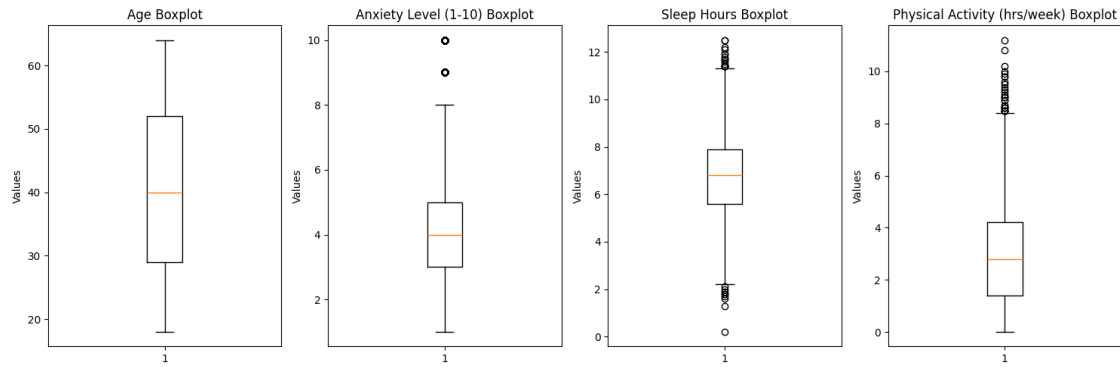
Max ZScore: 4.07

Outlier values: [11.8, 12.2, 11.7, 1.9, 12.5, 1.8, 1.7, 1.8, 1.6, 11.9, 11.9,  
1.3, 12.1, 11.7, 0.2, 12.5, 11.7]

Physical Activity (hrs/week): 30 outliers (0.27%)

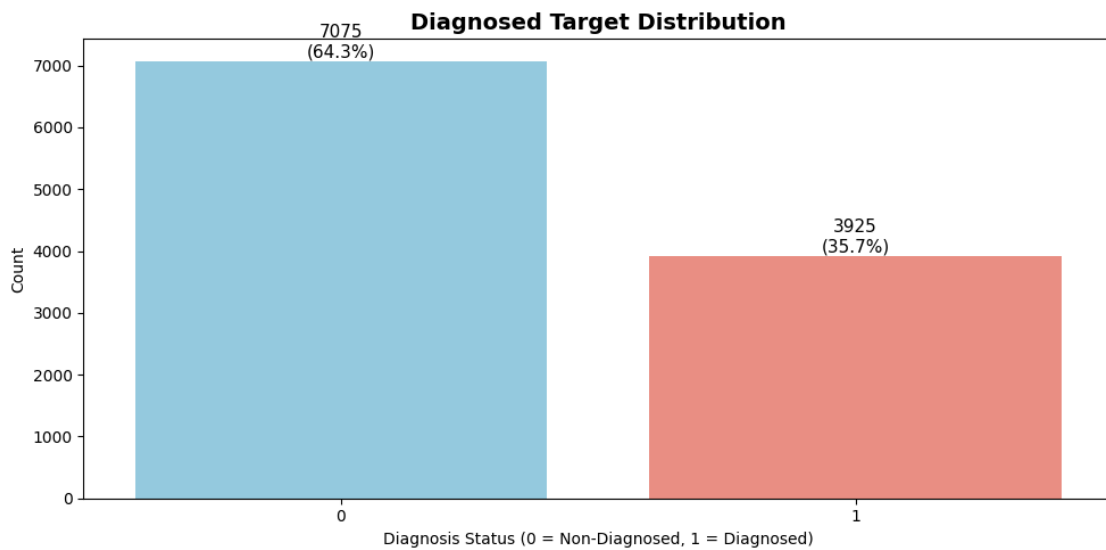
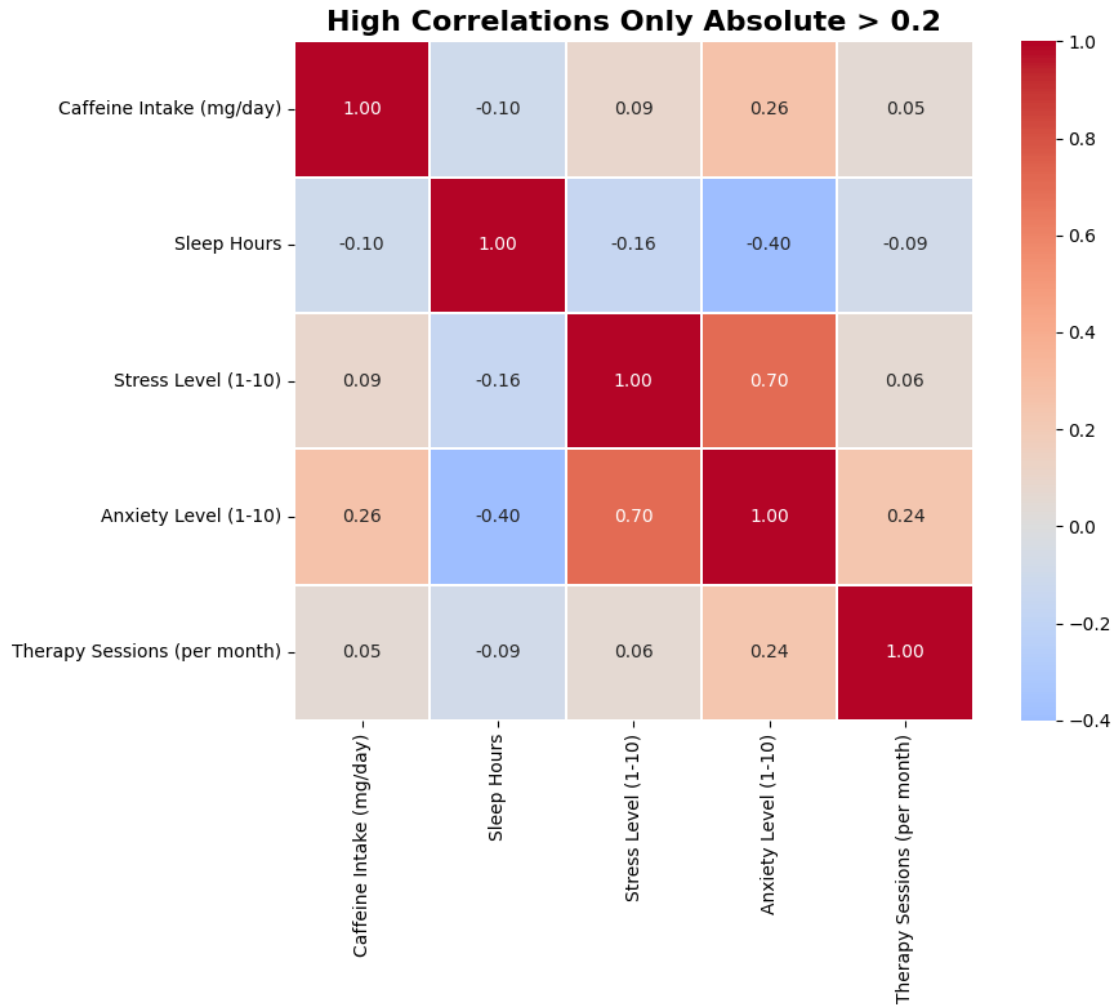
Max ZScore: 4.51

Outlier values: [10.8, 9.2, 8.5, 9.8, 10.2, 8.6, 8.5, 8.5, 10.0, 9.3, 8.6,  
8.7, 9.6, 8.5, 9.5, 11.2, 9.8, 8.5, 9.0, 8.5, 9.4, 8.7, 9.9, 8.6, 9.1, 8.6, 9.0,  
8.9, 9.0, 8.9]

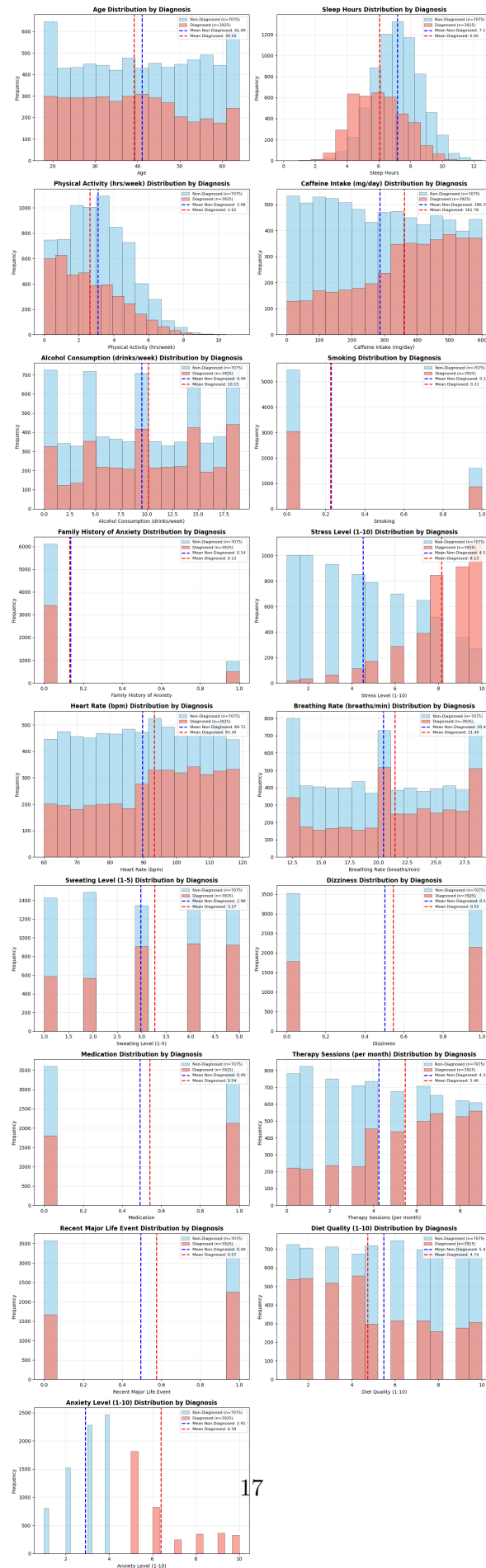


HIGH CORRELATIONS ABS > 0.2

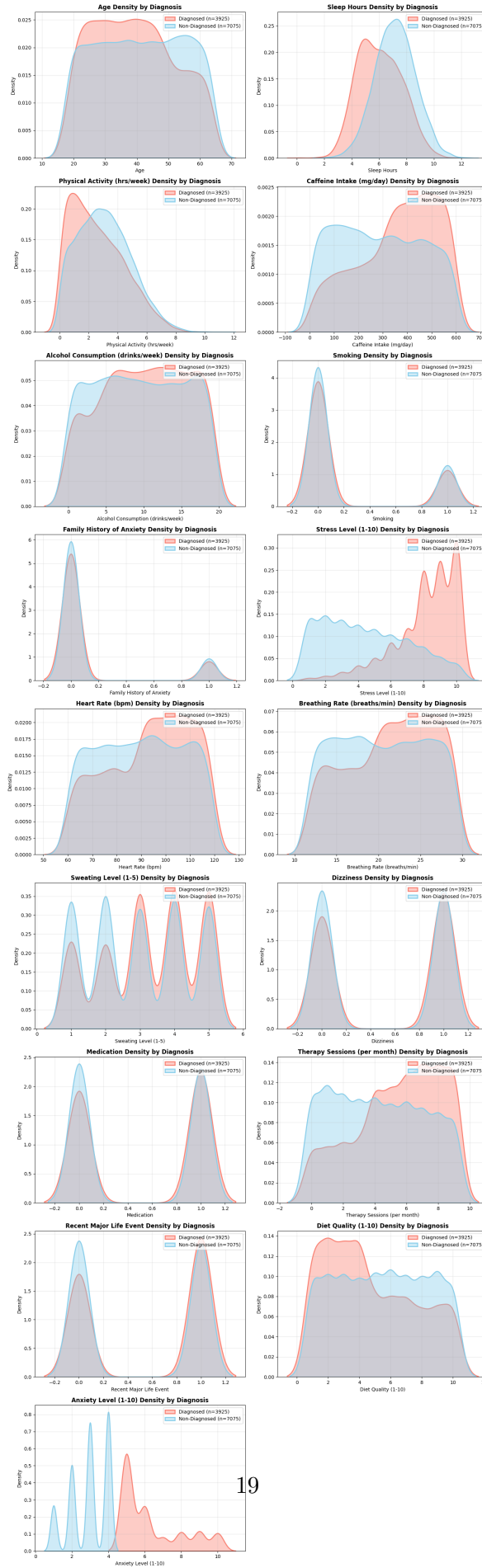
|  | Feature 1                    | Feature 2            | Correlation |
|--|------------------------------|----------------------|-------------|
|  | Stress Level (1-10)          | Anxiety Level (1-10) | 0.699490    |
|  | Caffeine Intake (mg/day)     | Anxiety Level (1-10) | 0.256051    |
|  | Therapy Sessions (per month) | Anxiety Level (1-10) | 0.240451    |
|  | Sleep Hours                  | Anxiety Level (1-10) | -0.401197   |











# SKEWNESS AND KURTOSIS OF FEATURES

|                                   | Skewness  | Kurtosis  |
|-----------------------------------|-----------|-----------|
| Family History of Anxiety         | 2.154874  | 2.643963  |
| Smoking                           | 1.308571  | -0.287695 |
| Anxiety Level (1-10)              | 0.918902  | 0.662642  |
| Physical Activity (hrs/week)      | 0.490496  | -0.272559 |
| Stress Level (1-10)               | -0.148151 | -1.273465 |
| Diet Quality (1-10)               | 0.139806  | -1.217157 |
| Heart Rate (bpm)                  | -0.128128 | -1.157126 |
| Caffeine Intake (mg/day)          | -0.115152 | -1.186871 |
| Therapy Sessions (per month)      | -0.099686 | -1.192781 |
| Recent Major Life Event           | -0.091380 | -1.992012 |
| Breathing Rate (breaths/min)      | -0.090613 | -1.199353 |
| Sweating Level (1-5)              | -0.088569 | -1.269781 |
| Dizziness                         | -0.071692 | -1.995223 |
| Age                               | 0.061239  | -1.156933 |
| Alcohol Consumption (drinks/week) | -0.040870 | -1.176549 |
| Medication                        | -0.035283 | -1.999119 |
| Sleep Hours                       | -0.011845 | -0.262601 |

## CATEGORICAL FEATURES SUMMARY

Categorical Feature: Gender

Number of unique values: 3

Value counts top 10:

Gender

Other 3706

Male 3666

Female 3628

Name: count, dtype: int64

Categorical Feature: Occupation

Number of unique values: 13

Value counts top 10:

Occupation

Scientist 880

Teacher 871

Artist 870

Engineer 868

Freelancer 864

Other 858

Lawyer 854

Athlete 847

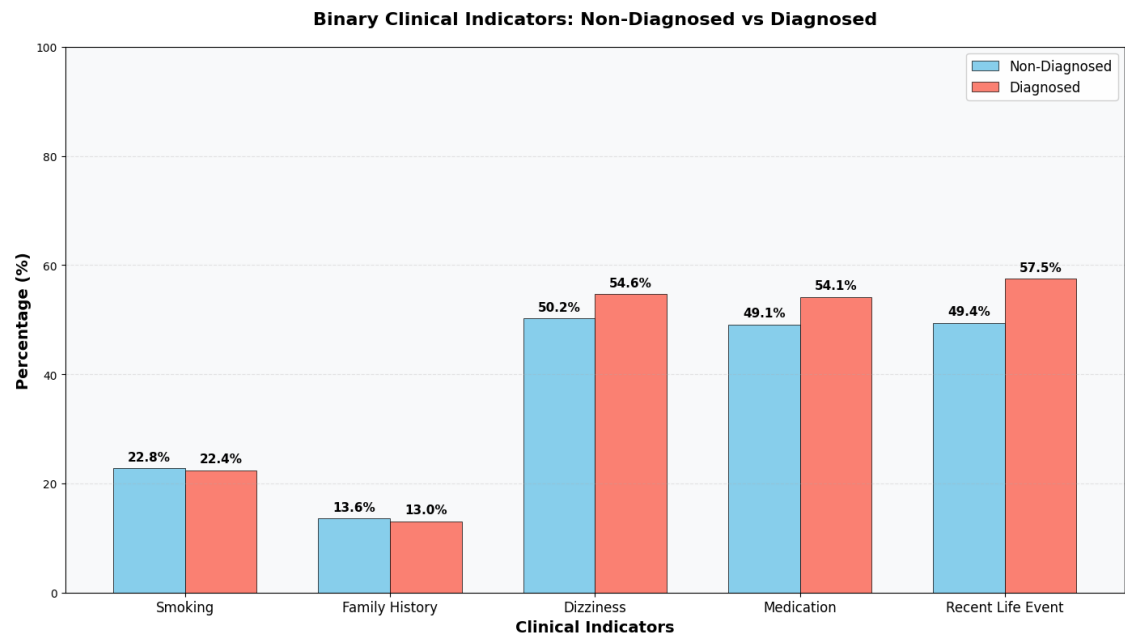
Nurse 833

Student 826

Name: count, dtype: int64

=====

BINARY FEATURES PERCENTAGES BY DIAGNOSIS GROUP  
=====



Binary Features - Percentageby Diagnosis Group:

|                   |                      |  |                  |  |                   |
|-------------------|----------------------|--|------------------|--|-------------------|
| Smoking           | Non-Diagnosed: 22.8% |  | Diagnosed: 22.4% |  | Difference: -0.4% |
| Family History    | Non-Diagnosed: 13.6% |  | Diagnosed: 13.0% |  | Difference: -0.6% |
| Dizziness         | Non-Diagnosed: 50.2% |  | Diagnosed: 54.6% |  | Difference: 4.4%  |
| Medication        | Non-Diagnosed: 49.1% |  | Diagnosed: 54.1% |  | Difference: 5.0%  |
| Recent Life Event | Non-Diagnosed: 49.4% |  | Diagnosed: 57.5% |  | Difference: 8.1%  |

Dataset saved  
Final dataset shape: 11000 rows, 20 columns